

AeroDefense is an expert company focused entirely on drone detection, and AirWarden is the brand name for its suite of drone detection systems. The company's core function is providing airspace awareness systems that detect, track, and locate both drones and their pilots, giving security teams the insight needed for quick and effective responses.

Here is a comprehensive overview of AeroDefense and its AirWarden solutions:

AeroDefense: Company Background and Mission

AeroDefense was **founded in 2015** with the mission to help organizations understand and manage drone activity in their airspace. The inspiration stemmed from 2015 incidents where drones interfered with firefighting efforts in California, crashed at the US tennis open stands, and landed near Angela Merkel.

- **Mission and Vision:** The company's goal is to provide **affordable, reliable, scalable drone detection** for any security-conscious entity. The mission is to equip organizations with the tools to understand and manage drone activity.
 - **Location and Manufacturing:** AeroDefense develops and manufactures its solutions in the US, headquartered in Oceanport, NJ. This US-based operation ensures data and hardware security.
 - **Core Approach:** The company began as "drone go home" but reformulated and rebranded as AeroDefense. It uses a legal detection method, passively detecting drone signals without interfering with or intercepting private drone communications, meaning all products **operate without the need for legal waivers** or special permissions.
 - **Customers and Deployments:** Since 2017, AeroDefense has served customers in **public safety, correctional facilities, stadiums, critical infrastructure, and defense sectors**. AirWarden systems have been commercially deployed since 2018 at MetLife Stadium and have proven effective in complicated environments like downtown Manhattan and Times Square during large outdoor gatherings.
 - **Achievements and Recognition:**
 - The company has secured **three Small Business Innovation Research (SBIR) Grant contracts** through the US Air Force to accelerate dual-use drone detection technology development.
 - AeroDefense holds **five US patents** and one in Canada.
 - AirWarden was the **first drone detection solution** to receive the Department of Homeland Security's (DHS) Support Anti-Terrorism by Fostering Effective Technologies (SAFETY) Act Development Testing and Evaluation Designation in 2019. DHS further awarded AirWarden full **Designation status in 2024**, which proves the system's effectiveness.
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AirWarden Drone Detection Systems

The AirWarden suite offers solutions designed to match a customer's investment to their level of risk. The systems provide **real-time tracking** and deliver airspace awareness, allowing security teams to determine the pilot's intent and request cooperation to safely land the drone if necessary.

1. AirWarden Essentials

AirWarden Essentials is a **cost-effective, wide-area drone and pilot detection solution**. It is specifically designed for quick deployment and affordability.

- **Technology:** Essentials leverages the **FAA Remote Identification (Remote ID) mandate**, which requires drones to broadcast identifying information and location (acting as a digital license plate). It primarily detects these Remote ID-compliant drones/signals.
- **Cost and Subscription:** The service is sold via flexible subscription plans, starting at **\$9,999 per year**. Subscribers typically pay for one year at a time.
- **Hardware and Installation:** The system involves a **small receiver unit**. The hardware is provided **complimentary** as part of the subscription service.
 - Installation is described as simple and quick, often taking only about **five minutes** to mount the receiver on a building or pole and plug it in. Full deployment can be completed in minutes.
 - The receiver connects automatically to the cloud-based command console over LTE or ethernet, and there is **no ongoing maintenance required**.
- **Use Cases:** Essentials is ideal for general businesses and government agencies, and can be used for both permanent and temporary deployments, such as securing airspace at **large public gatherings, concerts, or during natural disaster response**.

2. AirWarden Spectrum (or Classic)

For clients facing or suspecting more sophisticated threats, AeroDefense offers solutions designed for higher-risk areas.

- **AirWarden Spectrum** offers multiple layers of **advanced frequency-based** drone detection. It is designed to detect **both FAA Remote ID-compliant and non-compliant drones**.
- **AirWarden Classic** is described as a more advanced solution for detecting non-compliant or modified drones.

System Features and Data Analysis

The AirWarden platform is a secure, **cloud-based monitoring system** built on a Department of Defense's (DoD) Collaborative Drone Detection Network, enabling information sharing between users and organizations.

Key features include:

- **Real-Time Tracking and Alerts:** The intuitive user interface features a live detection map that displays and tracks real-time drone and pilot activity. Automated alerts are sent via text and email and include actionable intelligence, such as the **exact GPS locations of the drone and pilot**.
- **Historical Data & Reporting:** The platform allows users to analyze past drone incidents, including replaying flight paths and behavior. Historical data reports use charts, graphs, and heat maps to identify patterns and trends, aiding in evidence collection, training, and the development of proactive security measures. Reports are available in PDF and CSV formats.
- **Regional Monitoring:** The platform supports monitoring **multiple locations**—even across different states or regions—from a single interface.

Integration Capabilities

AirWarden is designed as an **open-platform** drone detection and airspace management solution with flexible architecture supporting interoperability through APIs, data sharing, and real-time synchronization.

AirWarden supports seamless integration with:

- **Uncrewed Traffic Management (UTM):** Integration enhances coordination between air traffic and drone operators by feeding real-time drone data into UTM networks.
- **Radar:** Compatibility with radar systems allows AirWarden to detect drones that may not emit electronic signatures, creating a robust solution for detecting even stealth or low-profile drones.
- **Cameras:** Camera systems can be integrated to begin recording automatically when a drone is detected.

Response to Drone Incidents (Recommended Protocol)

AeroDefense provides data to help security teams intervene effectively, noting that forcefully removing a drone (mitigation) is dangerous because its behavior is not guaranteed and it could injure people or damage property.

The recommended procedural countermeasure to unauthorized drone activity is:

1. **Send a security person** to the pilot's location.
2. **Inform the pilot** they are violating regulations and that the system has recorded their drone ID.
3. **Ask them to land the drone.**
4. If the pilot refuses, provide the system's data to **law enforcement or the FAA liaison**.

The pilot risks criminal charges, FAA fines for unsafe flying, and potential loss of their pilot's license. The data provided by AirWarden is essential for taking legal action against violations like stalking, espionage, or "peeping tom" activity.

